

Professor Knowledge System

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Introduction

The Professor Knowledge System is a Prolog and Python based application designed for this course to expedite academic data retrieval. It features an interactive command-line chatbot that makes it easy for users to ask questions about professors, course specifics and class times. The system successfully interprets structured knowledge bases by combining SWI-Prolog for logical reasoning with Python for user interaction and data processing.

Technologies Used

Python 3.11
Core application logic, chatbot interface, and automated document parsing.

SWI-Prolog
Knowledge representation engine for answering queries and logical rule execution.

PySwip
A smooth integration layer that connects SWI-Prolog with the Python runtime.

Libraries Used

PyYAML

Parses structured YAML professor files into dictionary objects.

Pandas & Openpyxl

Processes and extracts course schedules from the Excel spreadsheet.

Python-Docx

Extracts course descriptions and instructor notes from Word syllabus documents.

How It Works: The Pipeline

STEP 1

Data collection

Python parsers read source files including the YAML professors, Excel spreadsheet, and Word syllabus.

STEP 2

Knowledge Representation

Scripts automatically generate structured Prolog facts and logical rules stored in external `.pl` files.

STEP 3

Interactive Querying

PySwip bridges Python and Prolog, allowing users to query office hours, courses, days, times, schedules and professors by a chatbot.

Project End Goal

By converting unprocessed departmental papers into an intelligent Prolog knowledge bas
e, the Professor Knowledge System automates academic schedule lookups.
The program lowers manual administrative friction for instructors and students by combi
ning powerful Python parsers with a conversational query assistant.
In the end, the project's modular architecture, careful testing, and complete compliance
to the course requirements demonstrate high software engineering standards.

Example: Questions

Questions you can ask the system:

- Which courses does Anthony Adedoyin teach?
- Which courses does Appolo Tankeh teach?
- What days does CTEC 350.170 meet?
- What department does Anthony Adedoyin belong to?

Benefits

- Time Management

Instant access of office hours & schedules

- Data Accuracy

Reliable tracking of faculty course assignments

- Scalability

Simple addition of new academic departments